UNIVERSITY
OF JEDDAH

Assignment #1

College of Computer Science and Engineering

CCCN 321 : Embedded Systems

SUBMITTED BY:

Raghad Mohammed Alghamdi
ID NO. : 2111376

Introduction

This assignment aims to develop an automated car parking system using Arduino. The system utilizes sensors and LCD to detect and guide vehicles into designated parking spaces. By implementing this technology, we can enhance parking efficiency, reduce human errors, and optimize space utilization. In this report, we will discuss the design, implementation, and benefits of the car parking automation system.

Circuit and Hardware Description

Circuit Components:

1- Arduino UNO.

2- Breadboard.

3- resistor.

4- 2 IR sensor

5- Wires.

6- Servo Motor

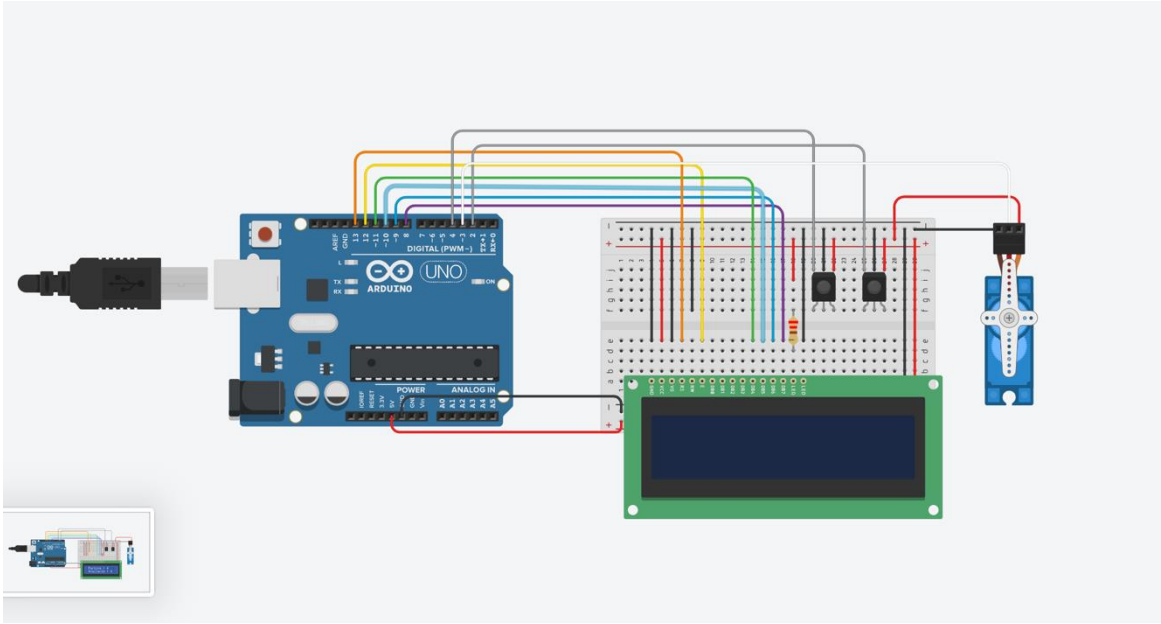
7- LCD 16X2

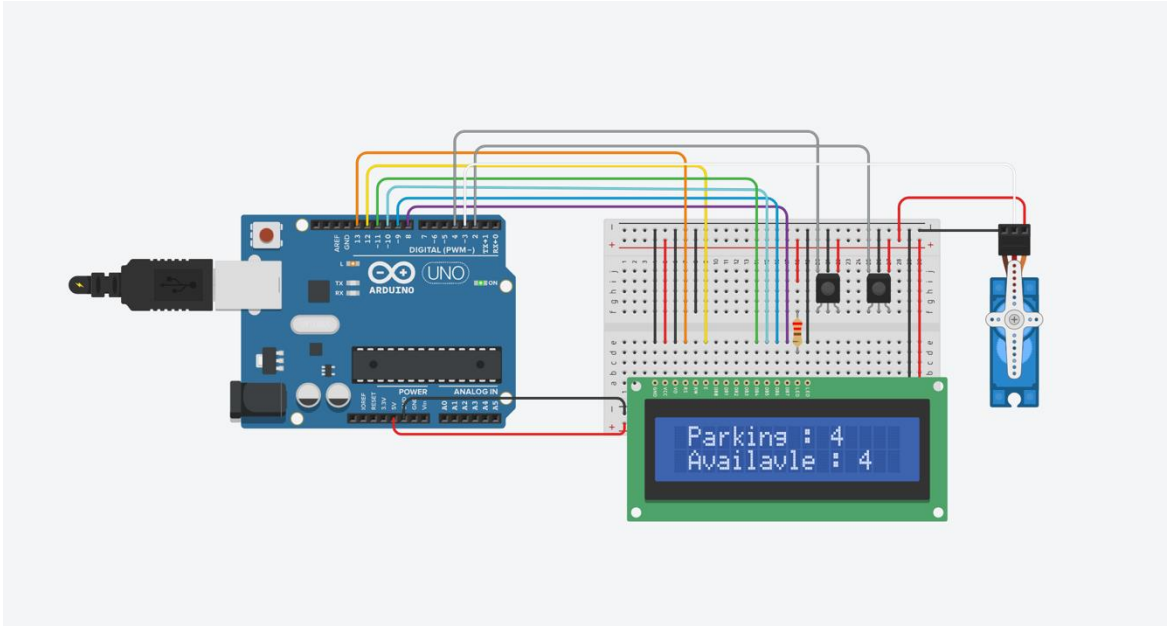
Circuit Wiring:

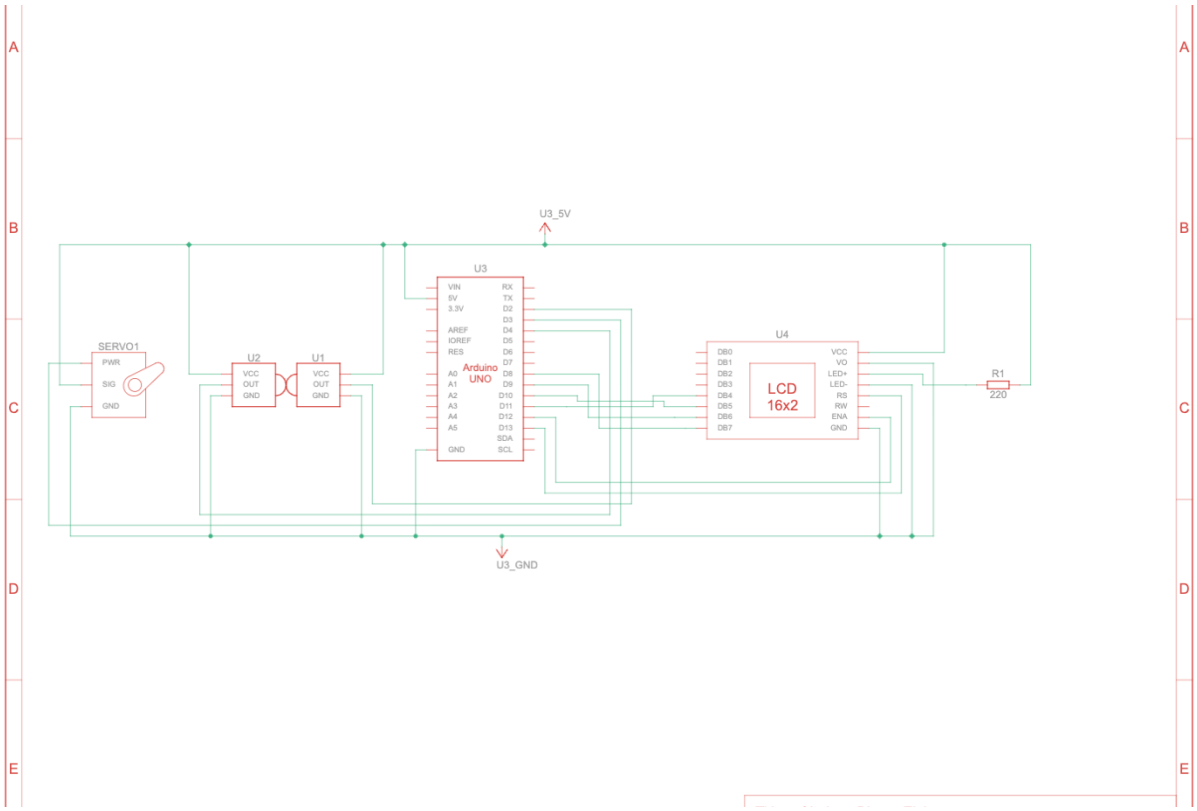
We will connect the sensors and the servo motor to the board and the Arduino, in addition to connecting the LCD as well. There will be 4 parking spaces here. When a car enters, the servo motor will open in addition to the decrease in the number of parking spaces, and this will appear on the LCD. When the number of parking spaces is complete, it will not open the Servo motor.

Circuit on software

Figure 1,2,3
Screenshot of the parking system Circuit on Tinkercad.

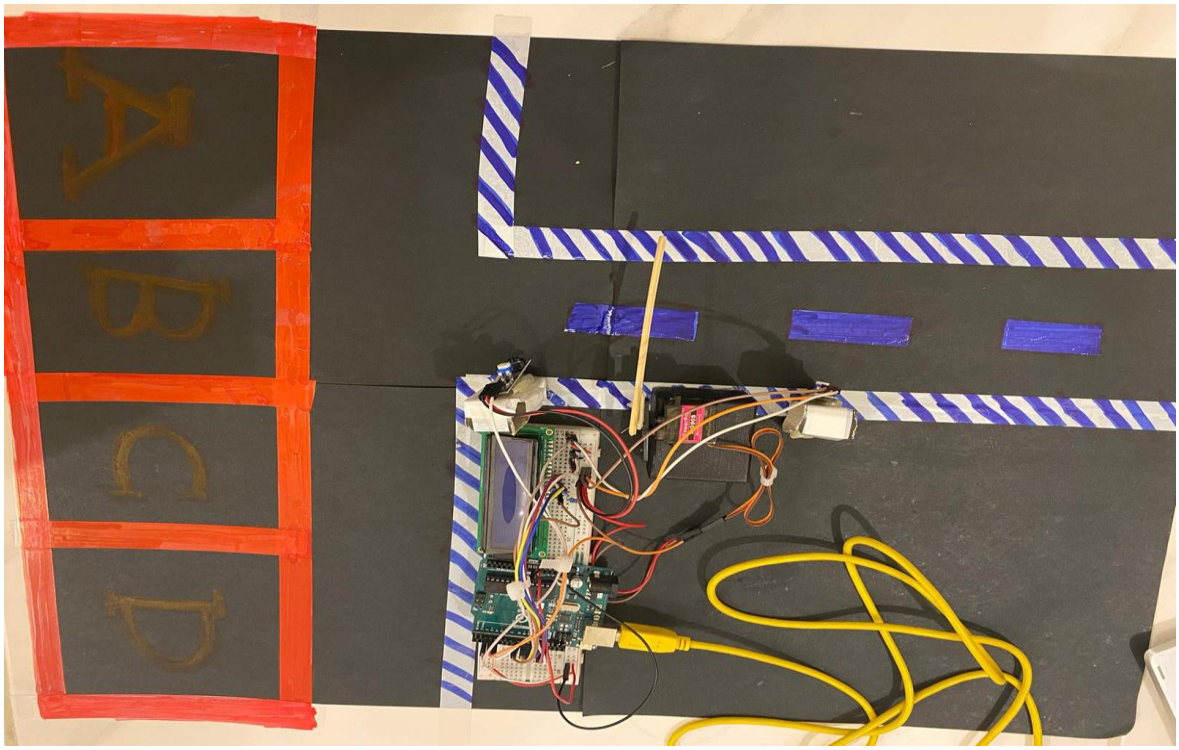


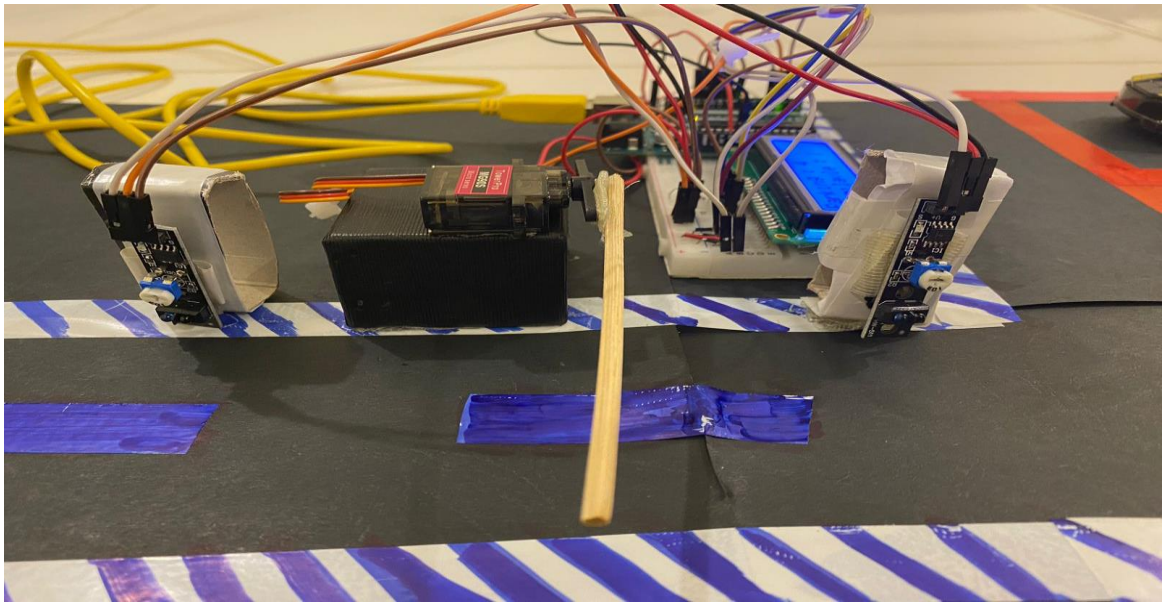



A
B
C
D
E
U3_5V
U3
VIN
5V
3.3V
AREF
IOREF
RES
A0
A1
A2
A3
A4
A5
GND
RX
TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
SDA
SCL
Arduino
UNO
SERVO1
PWR
SIG
GND
U2
VCC
OUT
GND
U1
VCC
OUT
GND
U4
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
VCC
VO
LED+
LED-
RS
RW
ENA
GND
LCD
16x2
R1
220
U3_GND

Circuit on Hardware

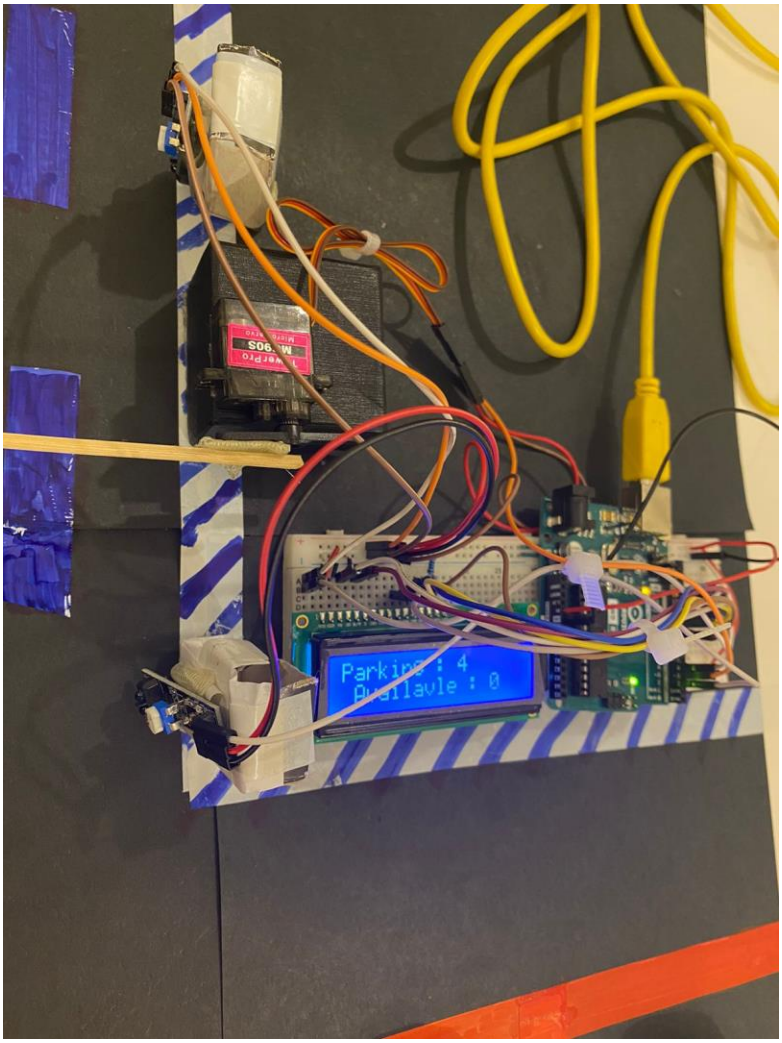
Figure 4,5
First Approach Circuit Wiring Physical





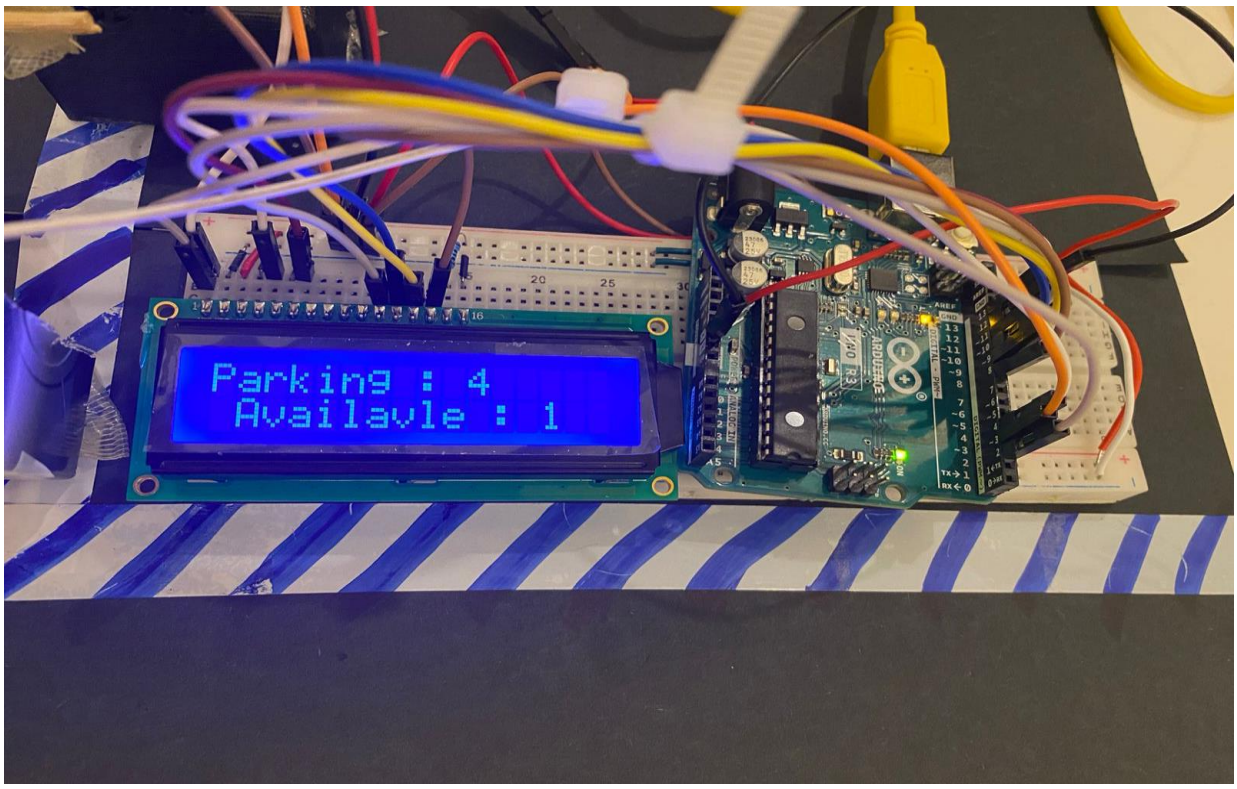
Circuit in Hardware

Figure 6
No parking are available.



Circuit on Hardware

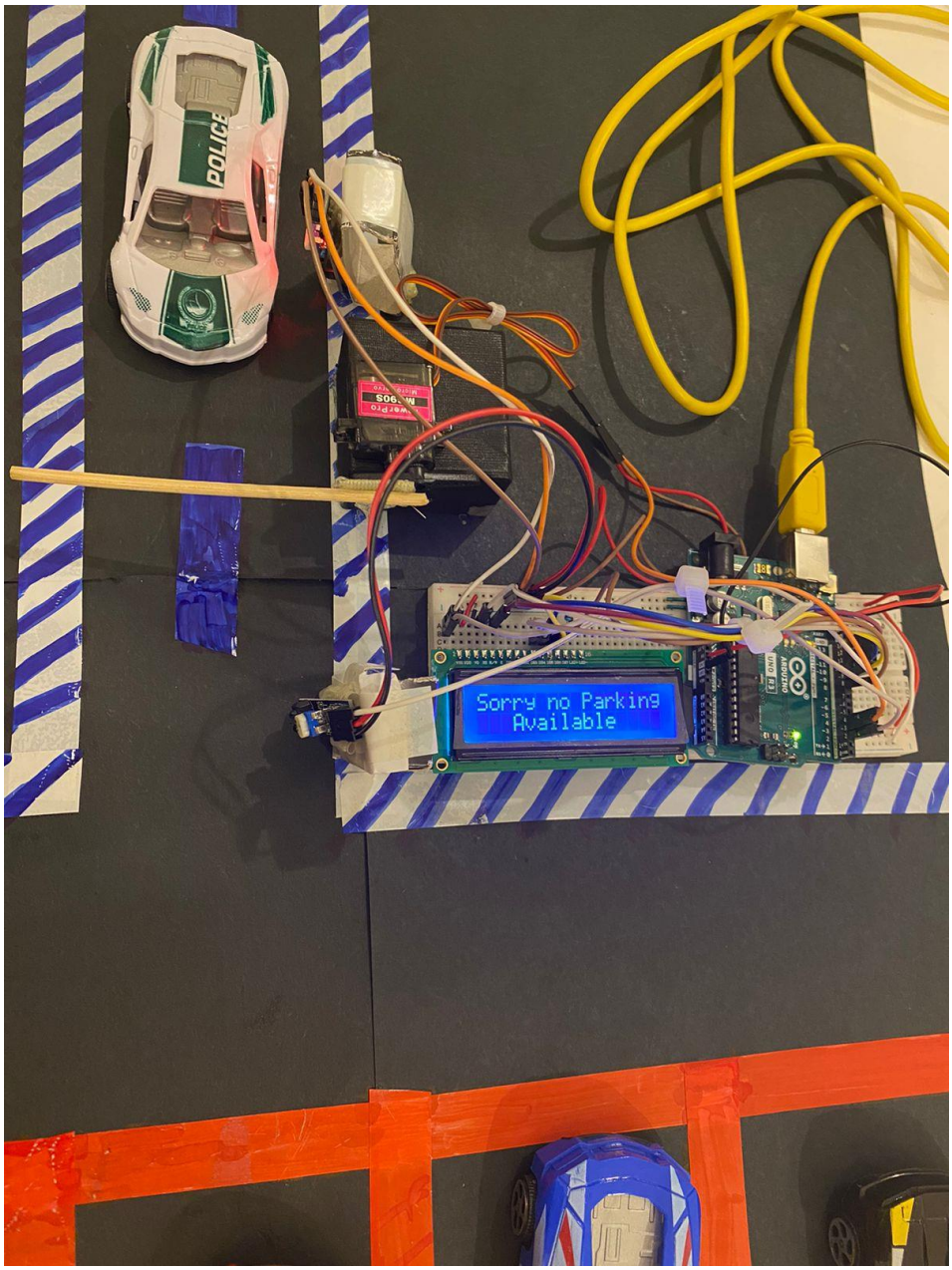
Figure 7
Only one parking are available.



Circuit on Hardware

Figure 8

Displaying a message that all parking is not available.



Code:

```
#include <LiquidCrystal.h>// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(13,12,11,10,9,8);// initialize the lcd input
#include <Servo.h> //includes the servo library

Servo myservo1;//declare the motor servo

int ir_s1 = 2;// initialize sensor 1
int ir_s2 = 4;// initialize sensor 2

int Total = 4;//number of parking
int Space;

int flag1 = 0;//initialize the flag1 to 0
int flag2 = 0;//initialize the flag2 to 0

void setup() {
pinMode(ir_s1, INPUT);//initialize the pin as input
pinMode(ir_s2, INPUT);//initialize the pin as input

myservo1.attach(3);//initialize the pin 3 in servo motor
myservo1.write(100);

lcd.begin(16, 2);// space massage in lcd
lcd.setCursor (0,0);
lcd.print("**Car  Parking**  ");//lcd message display
lcd.setCursor (0,1);//space massage in lcd
lcd.print("   *System*     ");//lcd message display
delay (2000);
lcd.clear();
```

```
Space = Total;
}
void loop(){
if(digitalRead (ir_s1) == LOW && flag1==0){
if(Space>0){flag1=1;
if(flag2==0){myservo1.write(0); Space = Space-1;}
}else{
lcd.setCursor (0,0);// space massage in lcd
lcd.print("Sorry no Parking "); //lcd message display
lcd.setCursor (0,1);//space massage in lcd
lcd.print("   Available    "); //lcd message display
delay (1000);
lcd.clear();
}
}

if(digitalRead (ir_s2) == LOW && flag2==0){flag2=1;
if(flag1==0){myservo1.write(0); Space = Space+1;}
}
if(flag1==1 && flag2==1){
delay (1000);
myservo1.write(100);
flag1=0, flag2=0;
}
lcd.setCursor (0,0);//space massage in lcd
lcd.print(" Parking : ");//lcd message display
lcd.print(Total);
lcd.setCursor (0,1);//space massage in lcd
lcd.print(" Availavle : ");//lcd message display
lcd.print(Space);
}
```

Conclusion

In conclusion, the parking system developed using Arduino provides an automated solution to parking issues and improvements. This system enhances parking efficiency, reduces human errors, and optimizes space utilization. It can be implemented in various applications such as commercial buildings, hospitals, and shopping centers. With the use of Arduino technology, sensors, and smart control, significant enhancements in the user experience of parking can be achieved.

References

https://www.tinkercad.com/things/lxZieFbm461-glorious-bigery-elzing

https://youtu.be/WGLaPoXJHkg?si=GDgsTz1A5MB4dfBO

https://youtu.be/6gccSyp_uJQ?si=cPWQ1vi3eFYZgAEh

https://www.tinkercad.com/things/7LCrckKtWNf